

Functional Description

Electronic Equipment

GenoLine NG

CANopen Data Protocol for Connecting Non-MTU Systems

E532463/00E



Power. Passion. Partnership.

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1 CANopen

1.1 CANopen for GenoLine NG

General information about the interface

This document describes the external CANopen interface for the GenoLine NG system. The document provides a general summary of the CANopen application.

CANopen protocol

The CANopen protocol is a standardized Level 7 protocol for CAN bus systems. The GenoLine NG system is always realized as a CANopen slave with the external side serving as the CANopen master.

Supported baud rate: 125 Kbaud

CANopen CAN-Frame

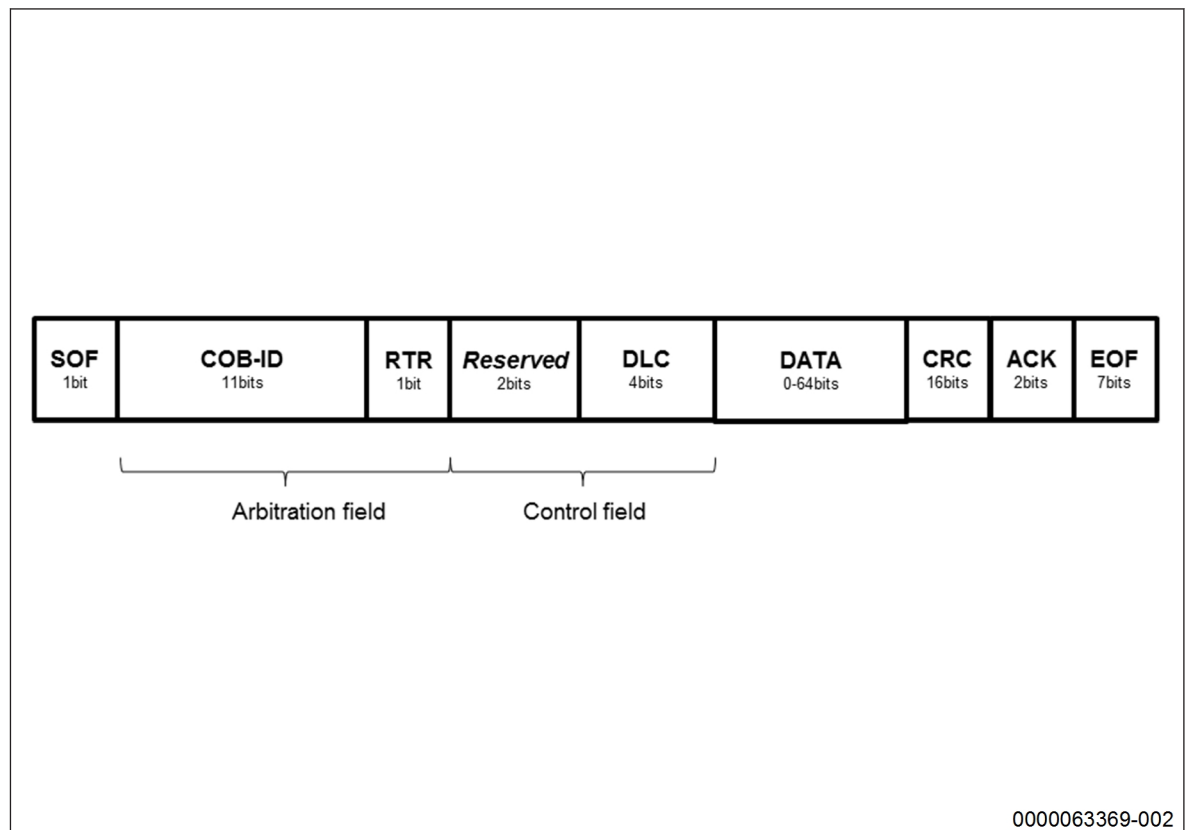


Figure 1: CAN frame

Item	Designation	Meaning
SOF	Start of Frame	Dominant bit starting a CAN frame
COB ID	CAN Object Identifier	Comprises 4 bits of function code and a 7 bit Node-ID
RTR	Remote Transmission Request	Always dominant in Standard Frames 2.0A
Re-served	-	Always dominant
DLC	Data Length Code	Length of the payload data
DATA	Payload data	MSB (Most Significant Bit) is transmitted first

TIM-ID: 0000077128 - 001

Item	Designation	Meaning
CRC	Cyclic Redundancy Check	Checksum used to detect data error
ACK	Acknowledgment	Dominant acknowledgment from individual CAN bus stations
EOF	End of Frame	Seven consecutive recessive bits

A standard CAN 2.0A frame with 11-bit identifier is used for CANopen.

1.2 CANopen state machine

Overview

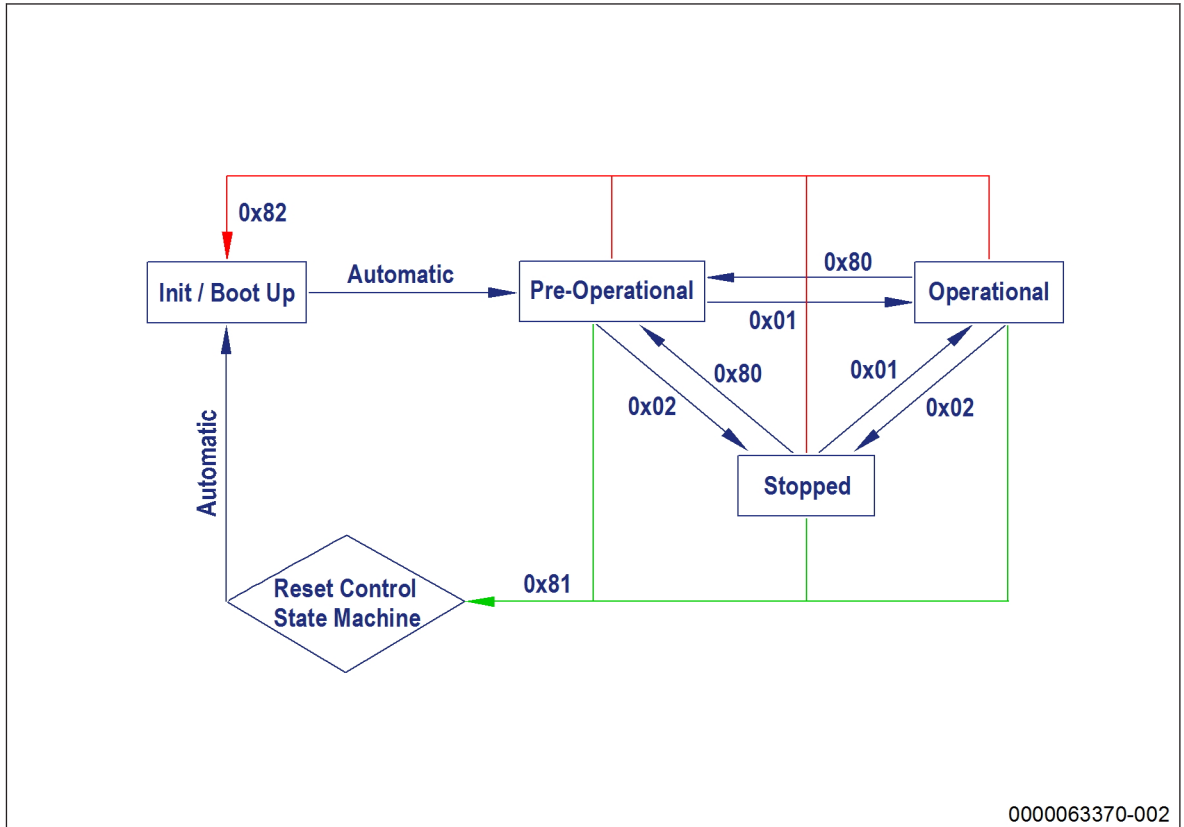


Figure 2: State machine states

Network management services allow CANopen master nodes to change various states of a CANopen slave node. These various states and transitions are illustrated in the figure.

The CANopen master can transmit the following status messages in the network:

Note: The DLC (Data Length Code) for status messages in the network is DLC = 2 bytes.

Action	CAN ID	Data byte 1	Data byte 2
Transition to Operational	0	0x01	Slave node ID (0 = all nodes)
Transition to Stopped	0	0x02	
Transition to Preoperational	0	0x80	
Reset CAN node	0	0x81	
Reset communication	0	0x82	

Table 1: Status messages in the network

State machine states

Initializing / Bootup

This is the default state of any slave node during initialization. Slave nodes automatically assume the “Preoperational” state after completing initialization. Slave nodes transmit a node-specific “Boot-up message” after successfully completing the boot sequence.

The state is attained by resetting the node or overall communication.

CAN-ID: 0x700 + Node, Data Byte = 0

Example: CANopen slave node 24 transmits a “Boot-up message”: CAN-ID: 0x718, Data Byte = 0

The CANopen master receives the following message:

Direction	ID (HEX)	Name	Interpretation	Data (HEX)
Rx Master	718	HBGuard_024	Boot-up	00

Table 2: Example: CANopen slave node 24 transmits a “Boot-up message”

Preoperational

PDO communication does not take place via the CANopen slave in this state. The CANopen master can change to the “Operational” state or to the “Stopped” state.

Note: The external interface to the GenoLine NG system does not facilitate SDO communication and does not provide an object directory. The emphasis is placed on the transmission of process data.

Example: CANopen master node 31 sets CANopen slave node 24 to “Operational” (0x01):

CAN-ID: 0x00, Data Byte1 = 0x01, Data Byte2 = 0x18

Direction	ID (HEX)	Name	Interpretation	Data (HEX)
Tx Master	0	NMTZeroMsg	Startnode, ID 18	01 18

Table 3: CANopen master node 31 sets CANopen slave node 24 to “Operational” (0x01)

Operational

The CANopen slave is able to perform PDO communication and transmit/receive process data in this state. The CANopen master can change to the “Preoperational” state or the “Stopped” state.

Example: CANopen master node 31 sets CANopen slave node 24 to “Stopped” (0x02):

CAN-ID: 0x00, Data Byte1 = 0x02, Data Byte2 = 0x1F

Direction	ID (HEX)	Name	Interpretation	Data (HEX)
Tx Master	0	NMTZeroMsg	Startnode, ID 18	02 18

Table 4: CANopen master node 31 sets CANopen slave node 24 to “Stopped” (0x02)

Stopped

The CANopen slave is completely disconnected. Neither PDO nor SDO communication are possible. Node Guarding or Heartbeat are possible to monitor the CAN node concerned. The CANopen master can change to the “Preoperational” state or the “Operational” state.

Example: CANopen master node 31 sets CANopen slave node 24 to “Preoperational” (0x80):

CAN-ID: 0x00, Data Byte1 = 0x80, Data Byte2 = 0x18

Direction	ID (HEX)	Name	Interpretation	Data (HEX)
Tx Master	0	NMTZeroMsg	Enter Preoperational, ID 18	80 18

Table 5: CANopen master node 31 sets CANopen slave node 24 to “Preoperational” (0x80)

1.3 Fault monitoring

“Fault monitoring” (not used)

The CANopen master polls the current status of the CANopen slave in the state machine via a Remote Frame. The CANopen master assumes that the CANopen slave has failed if the node fails to respond within a certain time. The CANopen master can then respond accordingly. The state is transmitted in the form of a byte. The highest data bit (7) constantly changes.

Note: Node guarding is not supported by the FMCS CANopen for GenoLine NG, heartbeat must be used in all cases.

“Heartbeat” failure detection

The CANopen slave transmits a periodic Heartbeat message indicating its present state. The CANopen master assumes that the CANopen slave has failed if this message is not received within a certain time. The CANopen master can then respond accordingly. The state is transmitted in the form of a byte. The highest data bit (7) must always be “0”.

Transmission for “Node Guarding” and “Heartbeat”

CAN-ID: 0x700 + Node:

- Data Byte = 0x00 (Boot-up)
- Data Byte = 0x04 (Stopped)
- Data Byte = 0x05 (Operational)
- Data Byte = 0x7F (Preoperational)

Note: The DLC (Data Length Code) for heartbeats in the network is DLC = 1 byte.

Example: CANopen slave node 24 transmits a heartbeat in the Preoperational state:

CAN-ID: 0x718, Data Byte = 0x7F:

Direction	ID (HEX)	Name	Interpretation	Data (HEX)
Rx Master	718	HBGuard_024	T0 Preoperational	7F

Table 6: Example: CANopen slave node 24 signals a heartbeat in the Preoperational state

Example: CANopen master node 31 transmits a heartbeat in the Operational state:

CAN-ID: 0x71F, Data Byte = 0x05

Direction	ID (HEX)	Name	Interpretation	Data (HEX)
Tx Master	71F	HBGuard_031	T0 Operational	05

Table 7: Example: CANopen master node 31 transmits a heartbeat in the Operational state

Note: The CANopen master node has to transmit a heartbeat to the CANopen slave. The slave has a 1000 ms timeout in this case. Otherwise PDO communication does not take place and the slave returns to the preoperational state. The NMT state of the master node has no relevance for the slave in this case.

1.4 Configuration for GenoLine NG

The CANopen master and CANopen slave node numbers have to be configured to suit the order-specific requirements of each genset.

Node number ranges: 1 to 127 (master and slave)

The CANopen master and CANopen slave node numbers must be different. Unsatisfactory communication between CANopen master and CANopen slave will result if the node numbers are configured incorrectly.

1.5 PDO communication

All process data is transmitted periodically and does not have to be explicitly requested by the CANopen master. A CANopen-“Predefined Connection Set” cannot be used. All PDOs are transmitted with an 8-byte DLC (Data Length Code).

Signal mapping

A textual drawing lists the mapping of the process data used in detail. This textual drawing contains the following information:

- CANopen slave Tx signals (measuring points, limits, binary signals and alarms)
- CANopen slave Rx signals from the master (control commands)

GenoLine NG Tx objects:

- Measuring points and limits: From CAN-ID 400 (0x190)
- Alarms and binary signals: From CAN-ID 700 (0x2BC)

GenoLine NG Rx objects:

- From CAN-ID 900 (0x384)

Meaning of the various columns (Rx, Tx)

Designation	Meaning
PDO Name	Generic PDO type name – measuring points, limits, alarms, binary signals
CAN ID	Each PDO has a dedicated CAN-ID.
Signal ID	The signal IDs are arranged in ascending order for each PDO from 1 upwards
Signal Name	Name of the signal
Bit Size	Size of the signal
Start Byte	Start byte within the PDO
Start Bit	Start bit within the start byte
MSC Unit and Scaling Factor	Scaling factor for the unit concerned

Scaling and value ranges

Measuring points and limits

Refer to the interface specification for details of the various CAN-IDs and their units of measure. Max. two 32-bit signed measuring points are assigned to each PDO.

All analog Tx objects (measuring points and limits) have to be divided in accordance with their unit of measure and with the associated scaling factor:

- Example: Engine Speed → Unit rpm, scaling 10:
 $0x00001f40 = 8000 \text{ (dec)} = 800.0 \text{ rpm}$
- Example: Negative temperature → Unit °C, scaling 100:
 $0xFFFFFDF3 = -525 \text{ (dec)} = -5.25 \text{ °C}$

Applicable value range = 0 bis FFFFFFFF

Missing Data (MD) = 0x7FFFFFFF

Sensor Defect Data (SD) = 0x80000001